KAIZEN MATHS

Lesson starter

Poisson Distribution

Rates and Contexts

Name

S1

Adjust lambda to match the interval in the question before calculating.

Calls arrive at an average rate of 5 per hour. Find the probability of exactly 4 calls in 45 minutes.

S2

Translate the words into a Poisson interval probability.

The number of defects on a roll of fabric follows $X \sim Po(4)$. Find $P(1 \leq X \leq 4)$.

S3

Use a complement for at least statements where possible.

Emails arrive at random with mean 3.2 per minute. Find the probability of at least 3 emails in a minute.

S4

Use a cumulative probability for at most statements.

A printer has an average of 4.1 faults per day. Find the probability of at most 2 faults in a day.

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STA-3014-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Poisson Distribution

Rates and Contexts

Name _____

EA1

Adjust lambda to match the interval in the question before calculating.

Calls arrive at an average rate of 3 per hour. Find the probability of exactly 4 calls in 45 minutes.

EA2

Translate the words into a Poisson interval probability.

The number of defects on a roll of fabric follows X ~ Po(5). Find P(3 ≤ X ≤ 7).

Show your working

KAIZEN MATHS

Exit ticket

Poisson Distribution

Rates and Contexts

Name _____

EB1

Use a complement for at least statements where possible.

Emails arrive at random with mean 3.2 per minute. Find the probability of at least 4 emails in a minute.

Show your working

KAIZEN MATHS

Exit ticket

Poisson Distribution

Rates and Contexts

Name _____

EA1

Adjust lambda to match the interval in the question before calculating.

Calls arrive at an average rate of 3 per hour. Find the probability of exactly 4 calls in 45 minutes.

EA2

Translate the words into a Poisson interval probability.

The number of defects on a roll of fabric follows X ~ Po(5). Find P(3 ≤ X ≤ 7).

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Show your working

KAIZEN MATHS

Exit ticket

Poisson Distribution

Rates and Contexts

Name _____

EB1

Use a complement for at least statements where possible.

Emails arrive at random with mean 3.2 per minute. Find the probability of at least 4 emails in a minute.

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Teacher answers & working

Poisson Distribution · Rates and Contexts

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S1

Adjust lambda to match the interval in the question before calculating.

Calls arrive at an average rate of 5 per hour. Find the probability of exactly 4 calls in 45 minutes.

Answer:

$P(X = 4) = 0.1938.$

Working:

First adjust the mean rate to match the time interval.

45 minutes = $\frac{45}{60}$ hours.

lambda = 5 x $\frac{45}{60}$ = 3.75.

Let X be the number of calls in 45 minutes, so X ~ Po(3.75).

$P(X = 4) = \frac{e^{-3.75}3.75^4}{4!} = 0.1938.$

S2

Translate the words into a Poisson interval probability.

The number of defects on a roll of fabric follows X ~ Po(4). Find P(1 ≤ X ≤ 4).

Answer:

$P(1 \leq X \leq 4) = 0.6105.$

Working:

Between 1 and 4 inclusive means add exact probabilities from 1 to 4.

$P(1 \leq X \leq 4) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4).$

$P(1 \leq X \leq 4) = 0.0733 + 0.1465 + 0.1954 + 0.1954.$

$P(1 \leq X \leq 4) = 0.6105.$

S3

Use a complement for at least statements where possible.

Emails arrive at random with mean 3.2 per minute. Find the probability of at least 3 emails in a minute.

Answer:

$P(X \geq 3) = 0.6201.$

Working:

Let X be the number of emails in one minute, so X ~ Po(3.2).

At least 3 means X ≥ 3.

Use the complement: P(X ≥ 3) = 1 − P(X ≤ 2).

$P(X \leq 2) = 0.0408 + 0.1304 + 0.2087 = 0.3799.$

$P(X \geq 3) = 1 - 0.3799 = 0.6201.$

Teacher answers & working

Poisson Distribution · Rates and Contexts

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S4

Use a cumulative probability for at most statements.

A printer has an average of 4.1 faults per day. Find the probability of at most 2 faults in a day.

Answer:

$P(X \leq 2) = 0.2238.$

Working:

Let X be the number of faults in a day, so X ~ Po(4.1).

At most 2 means X ≤ 2.

$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2).$

$P(X \leq 2) = 0.0166 + 0.0679 + 0.1393 = 0.2238.$

EA1

Adjust lambda to match the interval in the question before calculating.

Calls arrive at an average rate of 3 per hour. Find the probability of exactly 4 calls in 45 minutes.

Answer:

$P(X = 4) = 0.1126.$

Working:

First adjust the mean rate to match the time interval.

45 minutes = $\frac{45}{60}$ hours.

lambda = 3 x $\frac{45}{60}$ = 2.25.

Let X be the number of calls in 45 minutes, so X ~ Po(2.25).

$P(X = 4) = \frac{e^{-2.25}2.25^4}{4!} = 0.1126.$

EA2

Translate the words into a Poisson interval probability.

The number of defects on a roll of fabric follows X ~ Po(5). Find P(3 ≤ X ≤ 7).

Answer:

$P(3 \leq X \leq 7) = 0.7420.$

Working:

Between 3 and 7 inclusive means add exact probabilities from 3 to 7.

$P(3 \leq X \leq 7) = P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7).$

$P(3 \leq X \leq 7) = 0.1404 + 0.1755 + 0.1755 + 0.1462 + 0.1044.$

$P(3 \leq X \leq 7) = 0.7420.$

Teacher answers & working

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EB1

Use a complement for at least statements where possible.

Emails arrive at random with mean 3.2 per minute. Find the probability of at least 4 emails in a minute.

Answer:

$P(X \geq 4) = 0.3975$.

Working:

Let X be the number of emails in one minute, so X ~ Po(3.2).

At least 4 means X ≥ 4.

Use the complement: P(X ≥ 4) = 1 − P(X ≤ 3).

$P(X \leq 3) = 0.0408 + 0.1304 + 0.2087 + 0.2226 = 0.6025$.

$P(X \geq 4) = 1 - 0.6025 = 0.3975$.